

Mingxuan Liu

Center of Biomedical Imaging Research (CBIR), Tsinghua University, Beijing, P.R. China

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EDUCATION

PhD Student, School of Biomedical Engineering, Tsinghua University, China	2024.09 - Now
Advisor: Prof. Qiyuan Tian	
BEng Student, School of Biomedical Engineering, Tsinghua University, China	2021.09 - 2024.08
BEng Student, Department of Chemical Engineering, Tsinghua University, China	2019.09 - 2021.08

AWARDS AND HONORS

• ISMRM Workshop Trainee Stipend Award (Awarded: \$1500)	2026
• ISMRM Educational Stipend Award	2026
• Tsinghua CBIR Annual Retreat Oral Session 2nd Place	2026
• BEIHAI Summit Second Prize of Oral Presentation	2025
• National Scholarship, Ministry of Education, P.R. China	2025
• MICCAI autoPET IV Challenge 1st Place Winner	2025
• MICCAI VLM3D Challenge 2nd Place Winner	2025
• OHBM Merit Award (Awarded: \$850, Top 1.6%)	2025
• ISMRM Educational Stipend Award	2025
• National Scholarship, Ministry of Education, P.R. China	2024
• First Prize at National Biomedical Engineering Innovation Design Competition for College Students	2024
• Outstanding Graduates of Beijing, Beijing Municipal Administration Committee	2024
• Outstanding Graduates of Tsinghua University	2024
• ISMRM Magna Cum Laude Merit Award (Top 15%)	2024
• ISMRM Educational Stipend Award	2024
• IEEE ICCBD+AI Best Paper Award	2024
• Singapore AI Health Summit Poster Competition 3rd Place Winner (Awarded: \$3000)	2023
• National Scholarship, Ministry of Education, P.R. China	2023
• Scholarship for Academic Excellence, Tsinghua University	2022
• Scholarship for Science and Technology Innovation Excellence, Tsinghua University	2022
• Second Prize for Outstanding Projects of the Students Research Training Program, Tsinghua University	2022
• Scholarship for Academic Excellence, Tsinghua University	2020

SELECTED PUBLICATION

co-first author | * corresponding author

JOURNAL PAPERS

- **Liu M**[#], Liao Y[#], Li H, Zhu J, Yang H, Hao Y, Qu H^{*}, Tian Q^{*}. *Quality-label-free Fetal Brain MRI Quality Control Based on Image Orientation Recognition Uncertainty*. Medical Image Analysis, 2026. DOI: <https://doi.org/10.1016/j.media.2026.103994>

- **Liu M#**, Tang J#, Li H, Yong C, Qi J, Li S, Wang K, Gan J, Wang Y*, Chen H*. *Spiking-PhysFormer: Camera-Based Remote Photoplethysmography with Parallel Spike-driven Transformer*. Neural Networks, 2025. DOI: <https://doi.org/10.1016/j.neunet.2025.107128>
- **Liu M**, Jiao Y, Lu J, Chen H*. *Anomaly Detection for Medical Images Using Teacher-Student Model with Skip Connections and Multiscale Anomaly Consistency*. IEEE Transactions on Instrumentation and Measurement, 2024. DOI: <https://doi.org/10.1109/TIM.2024.3406792>
- Wu C#, Chen Y#, Du Y, Zong J, Dong J, **Liu M**, Qin F*, Peng Y, Fan J, Wang C. *Towards practical Alzheimer's Disease diagnosis: A lightweight and interpretable spiking neural model*. Biomedical Signal Processing and Control, 2026, 117:109595. DOI: <https://doi.org/10.1016/j.bspc.2026.109595>
- Zheng J, Li Z, Zhong W, Wang Z, Li Z, Yang H, **Liu M**, Cao X, Liao C, Salat DH, Huang SY, Tian Q. *Effects of diffusion MRI spatial resolution on human brain short-range association fiber reconstruction and structural connectivity estimation*. Imaging Neuroscience, 2025. DOI: <https://doi.org/10.1162/IMAG.a.1089>

SELECTED CONFERENCE PAPERS

- **Liu M#**, Li Y#, Zhu J#, Yang H, Huang Y, Li H, Chen Y, Bai X, Liao Y, Qu H, Tian Q*. *FetalExtract-LLM: Structured Information Extraction in Free-Text Fetal MRI Reports Based on Privacy-Ensuring Open-weights Large Language Models*. MICCAI Workshop on Perinatal, Preterm and Paediatric Image Analysis (PIPPI), 2025. (Oral) Link: https://doi.org/10.1007/978-3-032-05997-0_11
- Bai X#, **Liu M#**, Chen Y, Yang H, Tian Q*. *Chest-OMDL: Organ-specific Multidisease Detection and Localization in Chest Computed Tomography using Weakly Supervised Deep Learning from Free-text Radiology Report*. Medical Imaging with Deep Learning (MIDL), 2025. (Poster) Link: <https://openreview.net/forum?id=ns6nq592HX>
- Li H#, **Liu M#**, Bai X, Liao Y, Zheng J, Yang H, Li Z, Qu H, Tian Q*. *FetalCSR: Multi-input Attention Fusion Network for Neural ODE-based Fetal Cortical Surface Reconstruction*. ICLR Workshop on AI for Children (AI4CHL), 2025. (Oral) Link: <https://openreview.net/forum?id=Ra0xioC3He>
- Lin X#, **Liu M#**, Liu K, Chen H. *Spike-SLR: An Energy-efficient Parallel Spiking Transformer for Event-based Sign Language Recognition*. The British Machine Vision Conference (BMVC), 2024. (Poster) Link: <https://papers.bmvc2024.org/0493.pdf>
- Li S#, **Liu M#**, Zhang Y, Chen S, Li H, Chen H, Dou Z. *SAM-Deblur: Let Segment Anything Boost Image Deblurring*. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2024. (Poster) Link: <https://doi.org/10.1109/ICASSP48485.2024.10445844>
- **Liu M#**, Gan J#, Wen R#, Li T, Chen Y, Chen H*. *Spiking-Diffusion: Vector Quantized Discrete Diffusion Model with Spiking Neural Networks*. International Conference on Computer, Big Data and Artificial Intelligence (ICCBD+AI), 2024. (Best paper award) Link: <https://doi.org/10.1109/ICCBD-AI65562.2024.00110>
- **Liu M**, Jiao Y, Chen H*. *Skip-ST: Anomaly Detection for Medical Images Using Student-Teacher Network with Skip Connections*. IEEE International Symposium on Circuits and Systems (ISCAS), 2023. (Oral) Link: <https://doi.org/10.1109/ISCAS46773.2023.10181639>
- **Liu M**, Jiao Y, Gu H, Lu J, Chen H. *Data Augmentation Using Image-to-image Translation for Tongue Coating Thickness Classification with Imbalanced Data*. IEEE Biomedical Circuits and Systems Conference (BioCAS), 2022. (Oral) Link: <https://doi.org/10.1109/BioCAS54905.2022.9948645>

SELECTED CONFERENCE ABSTRACTS

- **Liu M**, Li H, Yang H, Liao Y, Zhu J, Zheng J, Hao Y, Li Z, Li Z, Wang Z, Li Y, Huang J, Ning G, Qu H, Tian Q*. *Cortical Brain in Utero Growth Among Fetuses with Germinal Matrix and Intraventricular Hemorrhage (GMH-IVH)*. ISMRM & ISMRT Annual Meeting & Exhibition, 2026. (Digital Poster) Link: <http://echo.ismr.org/abstracts/view/786064de-0ae1-469e-a161-a8d03618044b>
- **Liu M**, Hao Y, Liao Y, Li H, Zhu J, Yang H, Chen Y, Bai X, Qu H, Tian Q*. *Quality-Label-Free Stack-Level Quality Control Improves Fetal Brain Slice-to-Volume Reconstruction*. ISMRM Workshop on Unlocking

the Potential of Prenatal MRI, 2026. (Traditional Poster) Link: <https://echo.ismrm.org/abstracts/view/fedc0bc4-ed95-4361-a9c6-0ce061f610b3>

- **Liu M**, Li H, Li Z, Yang H, Zheng J, Qu H, Tian Q*. *Unsupervised Fetal Brain MRI Quality Assessment based on Orientation Prediction Uncertainty*. OHBM Annual Meeting, 2025. (Poster) [OHBM Merit Award, Top 1.6%] Link: <https://hal.science/hal-04974115>
- **Liu M**, Liao Y, Zhu J, Li H, Yang H, Zheng J, Li ZH, Li ZY, Qu H, Tian Q*. *FreeHemoSeg: Label-Free Deep Learning Framework for Automated Segmentation of Fetal Brain Germinal Matrix and Intraventricular Hemorrhage*. The Annual Meeting of ISMRM, 2025. (Digital Poster) Link: <https://archive.ismrm.org/2025/3790.html>
- **Liu M**, Li Y, Zhu J, Yang H, Huang Y, Li H, Chen Y, Bai X, Liao Y, Qu H, Tian Q*. *From Free Text to Usable Labels: Privacy-Ensuring Open-weights LLM-Enhanced Clinical Report Extraction for Fetal MRI*. Beijing-Tsinghua Health AI Summit (BEIHAI), 2025. (Oral) [Second Prize of Oral Presentation] Link: <https://hal.science/hal-05330462>
- **Liu M**, Li H, Zhu J, Zheng J, Yang H, Li Z, Li Z, Tian Q*. *Detecting Fetal Germinal Matrix and Intraventricular Hemorrhage in Brain MRI Using Label-free Deep Learning*. The Annual Meeting of the Radiological Society of North America (RSNA), 2024. (Oral) Link: <https://hal.science/hal-04961196>
- **Liu M**, Li H, Li Z, Yang H, Zheng J, Zhang X, Tian Q*. *Image Quality Assessment using an Orientation Recognition Network for Fetal MRI*. The Annual Meeting of ISMRM, 2024. (Oral Power Pitch) [Magna Cum Laude Merit Award, Top 15%] Link: <https://archive.ismrm.org/2024/0726.html>
- **Liu M**, Li H, Liao Y, Qu H, Tian Q*. *Label-free Image Quality Assessment of Fetal Brain MRI with Unsupervised Deep Learning*. Singapore AI Health Summit, 2023. (Poster Competition 3rd Place Winner, Top 3%) Link: <https://hal.science/hal-04961169>

SELECTED MANUSCRIPTS UNDER REVIEW

- **Liu M**#, Hao Y#, Liao Y#, Zhu J, Li H, Yang H, Chen Y, Li Y, Anmahapong K, Li Z, Zheng J, Kang M, Song Y, Lai H, Zhou X, Ning G, Qu H*, Tian Q*. *Annotation-free Deep Learning for Automated Detection and Segmentation of Fetal Germinal Matrix-Intraventricular Hemorrhage in Brain MRI*. The Lancet Medical Imaging & Theranostics. (Under Review)
- **Liu M**#, Bai X#, Zhang M, Chen Y, Yang H, Wang Z, Luo Y, Zhou Y, Han X*, Tian Q. *Preoperative CTA-based Deep Learning Model for Predicting AKI after TEVAR in Type B Aortic Dissection*. Radiology: Artificial Intelligence. (Submitting)
- Hao Y#, **Liu M**#, Zhu J#, Yang H, Li H, Ning G, Liao Y, Qu H*, Tian Q*. *PANDA: Patch-based Unsupervised Deep Learning for Brain Anomaly Detection via Age Prediction in Fetal MRI*. Imaging Neuroscience. (Under Review)
- Zong J#, Chen Y#, **Liu M**#, Du Y, Wu C, Wu B, Zhou G, Qin F*. *MUIT-TTA: Annotation-Free Intracranial Hemorrhage Segmentation via Pseudo-Anomaly Synthesis and Test-Time Adaptation*. Pattern Recognition. (Under Review)
- Hao Y#, Luo Y#, **Liu M**#, Yang H, Hu X, Zhu J, Anmahapong K, Li Y, Li Z, Chen Y, Sun N, Hu R, Kang M, Song Y, Lai H, Zhou X, Ning G, Liao Y, Qu H*, Tian Q*. *FetMRIE: Fetal Brain MRI Image Enhancement without Clean Reference*. Medical Image Analysis. (Submitting)
- Zhang Y#, **Liu M**#, Chen Y#, Wu Y*, Zhang L, Guo M, Yan Y, Wei R, Cui Z, Luo P, Chen F, Yang B, Han Y, Liu X, Tian Q*. *Multi-center Validation of Machine Learning Models for Predicting Myopia Progression across School-based Screening and Optical Intervention Settings in Chinese Children and Adolescents*. Advanced Intelligent Discovery. (Under Review)
- Yang H#, Liao Y#, **Liu M**, Zhu J, Li H, Jia F, Li Z, Zhang M, Zheng J, Wang Z, Zhang J, Chen Y, Li Y, Luo L, Liu J, Lai H, Zhou X, Kang M, Song Y, Ning G, Li Z, Zhang X, Qu H*, Tian Q*. *3D Fetal Brain Morphometry from Single Thick-slice MRI Stacks Using Deep Learning Super-Resolution*. Nature Biomedical Engineering. (Under Review)

- Li H#, **Liu M#**, Varadarajan D, Hu Z, Tian Q*, Polimeni JR*. *XSurfer: Reconstructing Surface Meshes of Cerebral and Cerebellar Cortex from Diverse MRI Data Using Untrained Neural Networks*. European Conference on Computer Vision (ECCV). (Under Review)
- Zhang X#, **Liu M#**, Chen Y, Zhu J, Anmahapong K, Huang Y, Zhang Y, Yang H, Liao Y, Ning G, Qu H, Tian Q*. *ASTAR: Automated Induction of Standardized Medical Radiology Reporting Templates from Large-Scale Clinical Free-Text Corpora*. International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI). (Under Review)
- Hu X#, **Liu M#**, Yang H#, Zhu J, Li Y, Chen Y, Li H, Li Z, Hao Y, Li Z, Zhang Y, Ning G, Liao Y, Qu H, Tian Q*. *INFANiTE: Implicit Neural Representation for High-Resolution Fetal Brain Spatio-Temporal Atlas Learning from Clinical Thick-Slice MRI*. International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI). (Under Review)
- Li Y#, **Liu M#**, Chang L#, Yang H, Anmahapong K, Wang Z, Hu X, Hao Y, Li H, Chen Y, Li X, Jia F, Liao Y, Qu H, Tian Q*. *Maximizing Domain Generalization in Automated Fetal Brain Biometry*. International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI). (Under Review)
- Li Z#, Xu Y#, Zhang L, Han T, Yang X, Wang Y, **Liu M**, Xin S, Liu L, Liao H*, Ning G*. *Democratizing Expert-Level Liver Sonography through an Autonomous Lightweight Ultrasound Robot*. Nature Biomedical Engineering. (Major Revision)
- Zhang M#, Yang H#, Xiao J, Liu W, Yin T, He Y, Zheng J, Li Z, **Liu M**, Liu W, Liao C, Bilgic B, Polimeni JR, Huang SY, Liao Y, Li Z*, Qu H*, Tian Q*. *SRNR: Deep Learning-based MRI Super-Resolution Using Noisy High-Resolution Reference Data*. Imaging Neuroscience. (Major Revision)
- Chen Y#, Zheng J#, Wang Y#, Wu B, Li L, **Liu M**, Xu L, Wu Y, Liu C, Guo L, Yang H, Bai X, Qin F, Liao Q, Gu Y, Zhao G, Ma L, Pan K, Guo J, Zhou Y, Sun H*, Tian Q*. *A Chinese Vision-Language Large Model for Emergency 3D Brain CT Interpretation*. npj Digital Medicine. (Minor Revision)
- Chen Y#, Li Z#, Wang Y, Li Y, Zheng J, Yang H, **Liu M**, Cukur T, Li Z, Tian Q*. *MicroKAN: Mapping Human Brain Microstructure Using Diffusion MRI and Convolutional Kolmogorov–Arnold Network*. NeuroImage. (Major Revision)

ACADEMIC SERVICE

- Conference Co-organizer: OCSMRM 2024, BEIHAI Summit 2025, MICS 2026.
- Journal Reviewer: Applied Intelligence, Biomedical Signal Processing and Control, Engineering Applications of Artificial Intelligence, IEEE Transactions on Cognitive and Developmental Systems, IEEE Transactions on Human-Machine Systems, Neural Networks, Neurocomputing, Pattern Recognition.
- Conference Reviewer: AAAI, ACM MM, ACM MobileHCI, BMVC, CHIL, ICASSP, ICME, IJCNN, ISBI, ISMRM, MICCAI, MIDL, NeurIPS, AAAI/ICLR/ICML Workshop.
- Teaching Assistant:
 - Medical Image Processing (Undergraduate Course, 3 Units), School of Biomedical Engineering, Tsinghua University, 2024/2025.
 - TIC3/4: Technological Innovation and Challenges 3/4, Weixian College, Tsinghua University, 2024/2025/2026.

INVITED TALKS

- 2026 ISMRM Workshop on Unlocking the Potential of Prenatal MRI Slides
- 2025 The 2nd Graduate Academic Forum (School of Biomedical Engineering, Tsinghua University) Slides
- 2025 Beijing-Tsinghua Health AI Summit (To Rupa Sarkar, Editor-in-Chief of The Lancet Digital Health) Slides
- 2025 MICCAI Workshop on Perinatal, Preterm and Paediatric Image Analysis (PIPPi) Slides
- 2024 Annual Meeting of Radiological Society of North America (RSNA) Slides
- 2024 ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM) Slides

- 2023 Singapore AI Health Summit Slides
- 2023 IEEE International Symposium on Circuits and Systems (ISCAS) Slides
- 2022 IEEE Biomedical Circuits and Systems Conference (BioCAS) Slides

SKILLS

**Programming & Python, PyTorch, TensorFlow, C/C++, Verilog, LATEX, MATLAB, Chisel, Arduino
Language & Mandarin Chinese (native speaker), English**